

# JWALIN SHAH

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AI Systems Engineer — agent reliability · evaluation · local inference

## PROFESSIONAL EXPERIENCE

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### Break the Web | AI Systems Engineer — LiveLM

May 2026 – Present

- Architecting LiveLM's grounding stack end-to-end: local retrieval over ~80K indexed canonical facts, REST/OpenAPI + MCP tool surfaces, and explicit ok / partial / stale / miss routing so the system falls back instead of inventing answers (~11 ms warm p50).
- Built the evaluation harness — 71 hand-labeled questions × 3 serving paths plus 512 story-anchored probes with sidecar checks on retrieval, freshness, answerability, and fallback.
- Current focus: ChatGPT Custom GPT integration against the live HybridSearch path and tuning serving-path behavior from benchmark + probe failures.

### tinyhumansai | Core Open-Source Contributor — OpenHuman

Mar 2026 – Present

- Landed 69 merged upstream PRs across a Rust/TypeScript desktop agent, spanning memory citations and namespaces, typed controller/RPC registries, privacy and security surfaces, chat reliability, and regression coverage.
- Shipped SQLite-backed contact identity resolution and scoring with CNContactStore; moved ~1,500 lines of CLI adapters into domain-owned Rust modules and added 196 tests across MCP, services, utilities, and autocomplete.
- Designed and prototyped an on-device personal index using FTS5 + sqlite-vec hybrid retrieval with staged iMessage, Gmail, Calendar, Contacts, and chronicle ingestion.

### Skild AI | Data Operations Lead — ML Systems & Reliability

San Mateo, CA · Jul–Nov 2025

- Built collection, validation, and evaluation systems across five robotic platforms; Python and Apps Script automation cut administrative overhead ~50% and improved downstream task success ~40%.
- Scaled a 30+ operator program with scheduling, QA, training, compliance, and runbooks; led technical execution for 25+ live demos during a \$1B Series C with zero demo failures.

## SELECTED SYSTEMS

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### Bridge — Reliable Coding-Agent Execution

Go · Seatbelt · Worktrees · Formal Methods

- Built an isolated spawn → verify → deliver runtime with allowlisted adapters, deny-default macOS sandboxing, leased worktrees, and fresh-checkout verification so workers cannot certify their own outputs.
- Cataloged 77 invariants and checked or proved 52 using 28 property-based checkers, 8 Z3 proofs, 5 TLA+ models (including a 1.58M-state exploration), and Lean proofs.

### Voice Engine — On-Device Speech Infrastructure

Swift · CoreML · MLX

- Fully local dictation (Moonshine ASR + VAD + punctuation + native injection). Benchmarked 26 ASR configs on 560 clips; chose moonshine-tiny (0.2177 WER, 0.05 RTF) over canary-qwen-2.5B (0.2162 WER, 0.73 RTF) — same accuracy class, **14× lower RTF**, real-time on device.
- 348/349 tests pass; one isolated macOS CoreML state regression documented. Harness isolates transcription, punctuation, vocabulary, and runtime failures.

### Jarvis & Inbox — Local-First Personal Agents

MLX · sqlite-vec · FastAPI

- Jarvis: privacy-first assistant on 8 GB M2 Air — 0.42 s mean / 1.15 s p95 fast path, Hit@5 0.88, 96.2% hallucination-gate pass across 37 configs; precursor patterns to OpenHuman memory work.
- Inbox: unified local TUI over iMessage, Gmail, Calendar, Notes, Reminders, GitHub, and Drive — the same personal sources later prototyped into OpenHuman life\_capture.

### OfficeQA Arena — Grounded Reasoning Evaluation

Python · Retrieval · Tool Use

- Reached 184.5/246 (75%) for \$1.71 after 12 architecture rounds and ~4,400 evaluations; evidence selection caused 48% of failures, while correctly grounded Python answers had 0% arithmetic errors.

## EDUCATION

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### University of Texas at Dallas — B.S. Computer Science, Dec 2024

Ultimate Frisbee — Captain · UTD Chess Team — member